

Bihar Engineering University, Patna

B.Tech. 1st Semester Examination, 2023

Course: B.Tech.

Time: 03 Hours

Code: 100103

Subject: Chemistry

Full Marks: 70

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Answer the following (Any seven question only):**[2 × 7 = 14]**

- (a) Why are Br^- and CCl_4 electrophiles?
- (b) What do you mean by single electrode potential?
- (c) Arrange molecular species N_2 , N_2^+ and N_2^- in increasing order of stability.
- (d) A water sample contains 204 mg of CaSO_4 per litre. Calculate the hardness in terms of CaCO_3 equivalent?
- (e) What is the value of effective nuclear charge (Z^*) for last electron of Cu^+ ? (Atomic no. of Cu = 29)
- (f) Out of Cr^{2+} and Cr^{3+} , which one is stable in aqueous solution?
- (g) What is the direction of a reaction, when $\Delta G = 0$?
- (h) How is the bond order related with dissociation energy?
- (i) The specific conductance of a decinormal solution of KCl at 18 °C is $0.0112 \text{ ohm}^{-1}\text{cm}^{-1}$. The resistance of the cell containing the solution at 18 °C was found to be 55 ohms. What is the cell constant?
- (j) A gas expands against vacuum. What is the work done on it?

Q.2 (a) Discuss Heisenberg's uncertainty principle and deduce a mathematical expression involving position and velocity of uncertainty of an electron. **[7]**

(b) Explain dual nature of light and give one example (property/experiment) in favour of its particle nature and wave nature. **[7]**

Q.3 (a) Draw the MO energy-level diagram for O_2 and based on the diagram explain the paramagnetic property observed in O_2 . **[7]**

(b) Explain ionization isomerism and linkage isomerism for transition metal complex with an example of each. **[7]**

Q.4 (a) 1 mole of an ideal gas expands isothermally and reversibly from 5 dm^3 to 10 dm^3 at 300 K. Calculate q, w, ΔU , ΔH , ΔG and ΔA . **[7]**

(b) Design electrochemical cells (in proper notation) in which each of the following reaction occurs: **[7]**

- (i) $\text{Ce}^{4+}(\text{aq}) + \text{Fe}^{2+}(\text{aq}) \rightarrow \text{Ce}^{3+}(\text{aq}) + \text{Fe}^{3+}(\text{aq})$
- (ii) $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$
- (iii) $\text{HgO}(\text{s}) + \text{H}_2(\text{g}) \rightarrow \text{Hg}(\text{l}) + \text{H}_2\text{O}(\text{l})$
- (iv) $\text{Fe}^{2+}(\text{aq}) + \text{Ag}(\text{s}) \rightarrow \text{Fe}^{3+}(\text{aq}) + \text{Ag}(\text{s})$

- Q.5** (a) Write the principle for lime soda process for softening of hard water? [7]
- (b) A water sample has the following constituents per litre. $\text{CaCO}_3 = 81 \text{ mg}$, $\text{Mg}(\text{HCO}_3)_2 = 75 \text{ mg}$, $\text{CaSO}_4 = 136 \text{ mg}$, $\text{MgSO}_4 = 120 \text{ mg}$, $\text{NaCl} = 4.7 \text{ mg}$. [7]
 Calculate the quantity of temporary and permanent hardness in the water sample.
 Calculate the quantity of lime (78% purity) and soda (92% purity) required for softening of 1.5 million litres of the water sample.
- Q.6** (a) The internuclear distance of NaCl is $2.36 \times 10^{-10} \text{ m}$. Calculate the reduced mass and moment of inertia of NaCl. (Atomic mass of Cl = $35 \times 10^{-3} \text{ kg mol}^{-1}$ and Na = $23 \times 10^{-3} \text{ kg mol}^{-1}$) [7]
- (b) Write the product for following reactions together with reaction mechanism: [7]
- (i) $2\text{CH}_3\text{COCH}_3 + \text{OH}^- \rightarrow$
- (ii) $\text{CH}_3\text{CH}_2\text{OH}$ (heated with H_2SO_4) $\rightarrow$
- Q.7** (a) Discuss photoelectric effect and explain equations involved in this effect. [7]
- (b) How many photons of light having a wavelength of $4000 \text{ \AA}$ are necessary to provide 1 J of energy? ($h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$) [7]
- Q.8** Explain the following: [3½ × 4 = 14]
- (a) SnI leads to racemic mixture, whereas Sn_2 gives rise to inverted product.
- (b) o-nitrophenol easily separates from p-nitrophenol by steam distillation.
- (c) The amino group in aniline is o- and p-directing, but amide group is m-directing.
- (d) Benzoic acid is weaker than formic acid.
- Q.9** (a) Write notes on the following: [3½ × 2 = 7]
- (i) Optical isomerism of lactic acid
- (ii) Diels – Alder reaction.
- (b) Differentiate between the following: [3½ × 2 = 7]
- (i) Enantiomers and diastereomers
- (ii) Racemic mixture and meso-compounds